INCH-POUND

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SUPERSEDING
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DETAIL SPECIFICATION

CORD ASSEMBLY, MILD DETONATION, CANOPY PERIPHERY

Inactive for new design after 7 May 2026.

This specification is approved for use by the Naval Surface Warfare Center Indian Head Division (NSWC IHD) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Canopy Periphery Mild Detonation Cord (MDC) Assembly, herein referred to as "MDC assembly".

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in sections 3 and 4 of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, users of this document are cautioned that they must meet all specified requirements of publications cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-8184 - Plastic Sheet, Acrylic, Modified

MIL-D-21625 - Design and Evaluation of Cartridges for Cartridge Actuated Devices

(Copies of these publications are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using ASSIST at https://assist.dla.mil.

AMSC N/A FSC 1377

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NAVAL AIR SYSTEMS COMMAND (NAVAIR) DRAWINGS

837AS118 - Cord Assembly, Mild Detonating, Canopy Periphery

837AS829 - Cord Assembly, Mild Detonating, Test Fixture

(Copies of these publications are available from Commander, Indian Head Division, Naval Surface Warfare Center, CAD/PAD Data Management Branch (Code E26), 4393 Benson Road, BLDG 1557, Indian Head, MD 20640-5092. Refer questions, inquires, or problems to nswc_iheodtd_dmo@navy.mil.)

2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E748 - Standard Guide for Thermal Neutron Radiography of Materials

ASTM E1742/E1742M - Standard Practice for Radiographic Examination

(Copies of these publications are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a first article sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Design and construction. The MDC assembly and its components shall be manufactured in accordance with Drawing 837AS118.

3.3 Detonation velocity. The detonation velocity of the detonating cord shall be 5,900 meters per second minimum when tested at -65 °F in accordance with 4.5.1. The detonation wave shall propagate through the entire length of the test specimen.

3.4 Core load. The range of the explosive cord loading in grains per foot, divided by the minimal core load, shall not exceed 24 percent when calculated from the detonating cord sample data obtained from 4.5.2.

3.5 MDC and booster interface.

3.5.1 Visual requirement. All MDC and booster interfaces shall be free of the following defects: inadequate or illegible descriptive marking or serial numbers, dents, deep scratches, or other defects that may prevent proper installation or function as specified in 4.5.3.

3.5.2 Leakage. All MDC and booster interfaces shall not exhibit a leak rate in excess of 1×10^{-5} cubic centimeters per second of air when tested as specified in 4.5.4.

3.5.3 Radiographic requirement. All MDC and booster interfaces shall be free of imperfections when examined radiographically as specified in 4.5.5.

3.5.4 Canopy periphery MDC assembly and subscale assembly.

3.5.4.1 Visual requirement. All MDC assemblies and subscale assemblies shall be free of the following defects: inadequate or illegible descriptive marking or serial numbers, dents, deep scratches, or other defects that may prevent proper installation or function as specified in 4.5.3.

3.5.4.2 Radiographic requirement. All MDC assemblies and subscale assemblies shall be free of imperfections when examined radiographically as specified in 4.5.5.

3.6 Environmental requirements.

3.6.1 Temperature, humidity, and altitude cycling. The MDC assembly shall not initiate or incur any degradation or damage when exposed to 28-day temperature, humidity, and altitude cycling test in accordance with MIL-D-21625 (see 4.5.6.1).

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3.6.2 Salt spray. The MDC assembly shall not initiate or incur any degradation or damage when exposed to salt spray in accordance with MIL-D-21625 (see 4.5.6.2).

3.7 Ballistic requirements. Subscale assemblies from each first article lot and production lot samples shall meet the requirements listed below when subjected to the tests specified in 4.5.7.

3.7.1 Performance. The MDC shall deliver sufficient energy to sever completely the 0.31±0.03-inch-thick cast acrylic plate in accordance with MIL-PRF-8184 over the temperature range of -65 to +200 °F.

3.7.2 Misfire. There shall be no misfires (see 4.5.7.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Production inspection (see 4.3).

c. Lot acceptance inspection (see 4.4).

4.2 First article inspection. Unless otherwise specified (see 6.2), an acceptable first article sample of 19 subscales shall be prepared using the same procedures and methods proposed for manufacture of the production lot. Eighteen of these subscales shall be expended in the tests listed in table I and one shall be retained for investigative purposes. An acceptable sample is one that has passed all the tests and examinations of the requirements in 3.3, 3.4, and 3.5. First article testing is for the purpose of determining that the contractor, prior to starting production, is capable of producing items that comply with the technical requirements as specified in the contract. As specified (see 6.2), a first article inspection is required if 2 or more years have passed since the item was manufactured, after major turnover in personnel, following major changes to the production facility, or following transfer of production to a new facility or site. Failure of any cartridge to comply with the requirements of table I shall be cause for rejection of the first article represented.

TABLE I. First article tests.

Test/Inspection	Requirement paragraph	Test paragraph	Quantity
Temperature, humidity, and altitude cycling	3.6.1	4.5.6.1	6
Salt spray	3.6.2	4.5.6.2	6
Ballistic (-65 °F)	3.7	4.5.7	3
Ballistic (+200 °F)	3.7	4.5.7	3

4.3 Production inspection. All MDC assemblies shall be inspected and screened to the examinations and tests listed in table II.

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TABLE II. Production inspection.

Item	Test	Requirement paragraph	Test paragraph
Detonating cord	Detonation velocity	3.3	4.5.1
Detonating cord	Core load	3.4	4.5.2
MDC and booster interface	Visual	3.2, 3.5.1	4.5.3
MDC and booster interface	Leakage	3.5.2	4.5.4
MDC and booster interface	Radiographic	3.5.3	4.5.5
MDC assembly	Visual	3.2, 3.5.4.1	4.5.3
MDC assembly	Radiographic	3.5.4.2	4.5.5

4.4 Lot acceptance inspection. Lot acceptance shall consist of the examinations and tests specified in table III. Failure of any sample subscale to comply with the requirements of table III shall be cause for rejection of the lot represented.

TABLE III. Lot acceptance inspection.

Test	Requirement paragraph	Test paragraph
Visual	3.5.4.1	4.5.3
Radiographic	3.5.4.2	4.5.5
Ballistic (-65 °F)	3.7	4.5.7
Ballistic (200 °F)	3.7	4.5.7

4.4.1 Sampling.

4.4.1.1 Detonating cord sampling. From each spool of detonating cord used in the production lot, a sample of 12-inch length of cord from each end of the spool and a 12-inch length every 100 feet of the cord shall be removed from the spool for testing. The 12-inch sample shall be divided into a 9-inch length for velocity tests and a 3-inch sample for the cord loading tests. Failure of any sample to meet the requirements of 3.3 and 3.4 shall result in the rejection of the spool of cord.

4.4.1.2 Subscale sampling. A sample of eight subscales shall be prepared from a random sample of the components used in the production lot. The lot size shall be limited to 91 maximum.

4.5 Examinations and tests.

4.5.1 Detonation velocity. The samples shall be tested for velocity measurements by detonating a 9-inch segment, conditioned at -65 °F for 3 hours minimum, with a No. 6 blasting cap with velocity measured over a fixed interval prior to producing any MDC assemblies. The velocity shall meet the requirements of 3.3.

4.5.2 Core load. The amount of core explosive for the 3-inch sample shall be determined to verify conformance with 3.4.

4.5.3 Visual inspection. All MDC and booster interfaces shall be visually examined prior to final assembly to determine conformance with 3.2 and 3.5.1. Then all MDC assemblies and subscale assemblies shall be examined to determine conformance with 3.5.4.1. Those with inadequate or illegible descriptive marking or serial numbers, dents, deep scratches, or other defects that may prevent proper installation or function shall be rejected.

4.5.4 Leakage. Each length of MDC and booster interfaces shall be leak tested in a dry gas leak tester to determine conformance with 3.5.2. Units that exhibit a leak rate in excess of 1×10^{-5} cubic centimeters per second of air at a pressure differential of 1.0 atmospheres shall be rejected.

4.5.5 Radiographic inspection. Each length of MDC and booster interface, prior to assembling in the canopy periphery, and each subscale shall be subjected to radiographic inspection in accordance with

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ASTM E1742/E1742M and ASTM E748. The samples shall be arrayed on trays or boards in consecutive numerical order and each radiograph shall carry a permanent identification of the MDC and subassembly displayed thereon. The MDC and subscales shall be positioned to give the most revealing cross-section of each unit. All units shall be identified with serial numbers prior to radiographic inspection. The radiographic identification shall include the complete lot number and the span of serial numbers displayed. Discontinuities in serial numbers shall be identified on the plate. Defective MDC and subscales shall be clearly identified on the plate and shall be removed. Radiographs or copies of the entire production lot shall accompany the ballistic sample to the activity conducting the production conformance test (see 6.2).

4.5.5.1 X-ray inspection. Each MDC length and booster interface and subscale shall be X-rayed in accordance with ASTM E1742/E1742M to verify the presence, condition, and proper installation of parts. Any observable imperfection shall be cause for rejection of the unit.

4.5.5.2 N-ray inspection. Each MDC and booster interface shall be N-rayed in accordance with ASTM E748 along the full length of the longitudinal axis to verify the presence of explosive, the absence of foreign hydrogenous material, and to verify that gaps and compaction densities are within control standards. Any observable imperfection shall be cause for rejection of the unit.

4.5.5.3 Loaded booster sleeves. Loaded booster sleeves shall be N-rayed separately and serialized. A marriage log shall be maintained that records each booster with the serial number of the MDC assembly for future reference.

4.5.6 Environmental tests.

4.5.6.1 Temperature, humidity, and altitude cycling. The MDC shall be subjected to a 28-day temperature, humidity, and altitude cycling test in accordance with MIL-D-21625 to verify conformance with 3.6.1.

4.5.6.2 Salt spray. The MDC shall be subjected to salt spray tests in accordance with MIL-D-21625 to verify conformance with 3.6.2.

4.5.7 Ballistic tests. Subscale assembly test samples shall be selected in accordance with 4.4.1.1. The samples shall be divided equally between the -65 °F and the +200 °F conditioning groups. The units shall be conditioned for a time period of not less than 4 hours or more than 24 hours. Subscales shall be fired as shown on Drawing 837AS829. Units shall be fired within 5 minutes of removal from conditioning. If not fired within 5 minutes, recondition for 2 hours and retest.

4.5.7.1 Acceptance criteria. The lot shall be acceptable with respect to functional testing if the ballistic sample meets the requirements of 3.7.1. Any failures or defects shall be cause for rejection of the lot represented.

4.5.7.2 Misfire. Any subscale that does not fire under the conditions specified in 4.5.7 shall be considered unacceptable, unless the misfire is clearly attributable to faulty test equipment or procedure.

4.5.7.3 Retest. There shall be no retests except as defined in 4.5.7 and 4.5.8.

4.5.8 Test failure. If a test failure is attributed to an assignable cause, excluding the test sample, the original test results shall be discarded and that test reconducted.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The MDC covered by this specification is for use in the emergency egress from the AV-8B aircraft. The Canopy Periphery Mild Detonation Cord (MDC) Assembly is designed to completely sever the aircraft canopy 360 degrees along the edge of the canopy in preparation for emergency egress from the aircraft.

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This explosive, in conjunction with the overhead MDC, fragmentizes the canopy allowing easier seat egress. This MDC is military unique and there is no equivalent commercial application for this design.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When first article inspection is required and, if so, specify the test activity (see 3.1 and 4.2).

c. Production lot size, sample size, and test activity.

d. Items of data required for each production lot.

e. Packaging requirements (see 5.1 and 6.6).

f. Safety requirements (see 6.5).

6.3 Flight critical part. The MDC is a flight critical item and the technical data package should be identified and procured as such.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1). The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles.

6.4.1 Bids. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product that has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Ammunition and explosive safety. (see 6.2)

6.5.1 Contractor's responsibility. The safety requirements of DoD Manual 4145.26, DoD Contractor's Safety Manual for Ammunition and Explosives, are applicable when this specification is used for acquisition.

6.5.2 Government activity's responsibility. The safety precaution requirements of OP 5, Ammunition and Explosives Safety Ashore, are applicable when this specification is used as part of a description of work to be accomplished by a government activity.

6.6 Contract packaging, packing and marking. The following requirements should be specified in the contract (see 6.2).

6.6.1 Packaging and packing. In addition to the following, level A and level C packaging and packing should be in accordance with 49 CFR 100-199. For explanation of terms, see 6.7.

6.6.1.1 Level A. Level A packaging and packing should be used for packaging and packing of all production lots for service use.

6.6.1.1.1 Inner container. A maximum of one assembly should be tightly packaged in KIMPAK dunnage, wrapped in a velostat bag, evacuated of air and heat-sealed. The assembly will then be placed in a cardboard container with enough dry dunnage to provide a tight pack.

6.6.1.1.2 Outer container. One inner container should be packed in the outer container conforming to ASTM D6880/D6880M, class 2, style 4 or 4½ or a box conforming to ASTM D6251/D6251M, overseas type, style A or B.

6.6.1.1.3 Level C. Minimum packaging and packing is required for shipping subscales, but a tight pack is required.

6.6.2 Marking.

6.6.2.1 Special marking. Marking of outer containers should be in accordance with CFR 49 171-199.

6.6.2.2 Normal marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages and shipping containers should be marked in accordance with MIL-STD-129.

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Inner container marking:--

NSN: 1377-01-136-7571 DODIC: MG40

Nomenclature: Cord Assembly, Mild Detonating, Canopy Periphery

Part Number: 30003-837AS118

Lot Number: (In accordance with MIL-STD-1168)

ICC Explosive Class: C

Outer container marking:--

NSN: 1377-00-136-7571 DODIC: MG40

Nomenclature: Cord Assembly, Mild Detonating, Canopy Periphery

Part Number: 30003-837AS118

Quantity: 1

Gross weight and cube:

Lot Number: (In accordance with MIL-STD-1168)

Contract or Purchase Order Number:

"WARNING: EXPLOSIVE POWER DEVICE, CLASS C EXPLOSIVE, HANDLE WITH CARE, KEEP FIRE AWAY"

The above information should be applied in accordance with the provisions of MIL-STD-129.

6.7 Definitions.

6.7.1 Level A packaging. The degree of preservation and packaging that will afford protection against corrosion, deterioration, and physical damage during shipment, handling, indeterminate storage, and worldwide redistribution.

6.7.2 Level C packaging. The degree of preservation and packaging that will afford protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to contractors' commercial practice when such meets the requirements of this level.

6.7.3 Level A packing. The degree of packing that will afford protection during shipment, handling, indeterminate storage, and worldwide distribution.

6.7.4 Level C packing. The degree of packing that will afford protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to contractors' commercial practice when such meets the requirements of this level.

6.8 Explanation of terms. For the purposes of this specification, NAVSUPINST 4030.55 should be used to clarify terms.

6.9 Subject term (key word) listing.

Aircraft canopy

Emergency egress

Mild detonating cord

Shape charge

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Preparing activity:
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